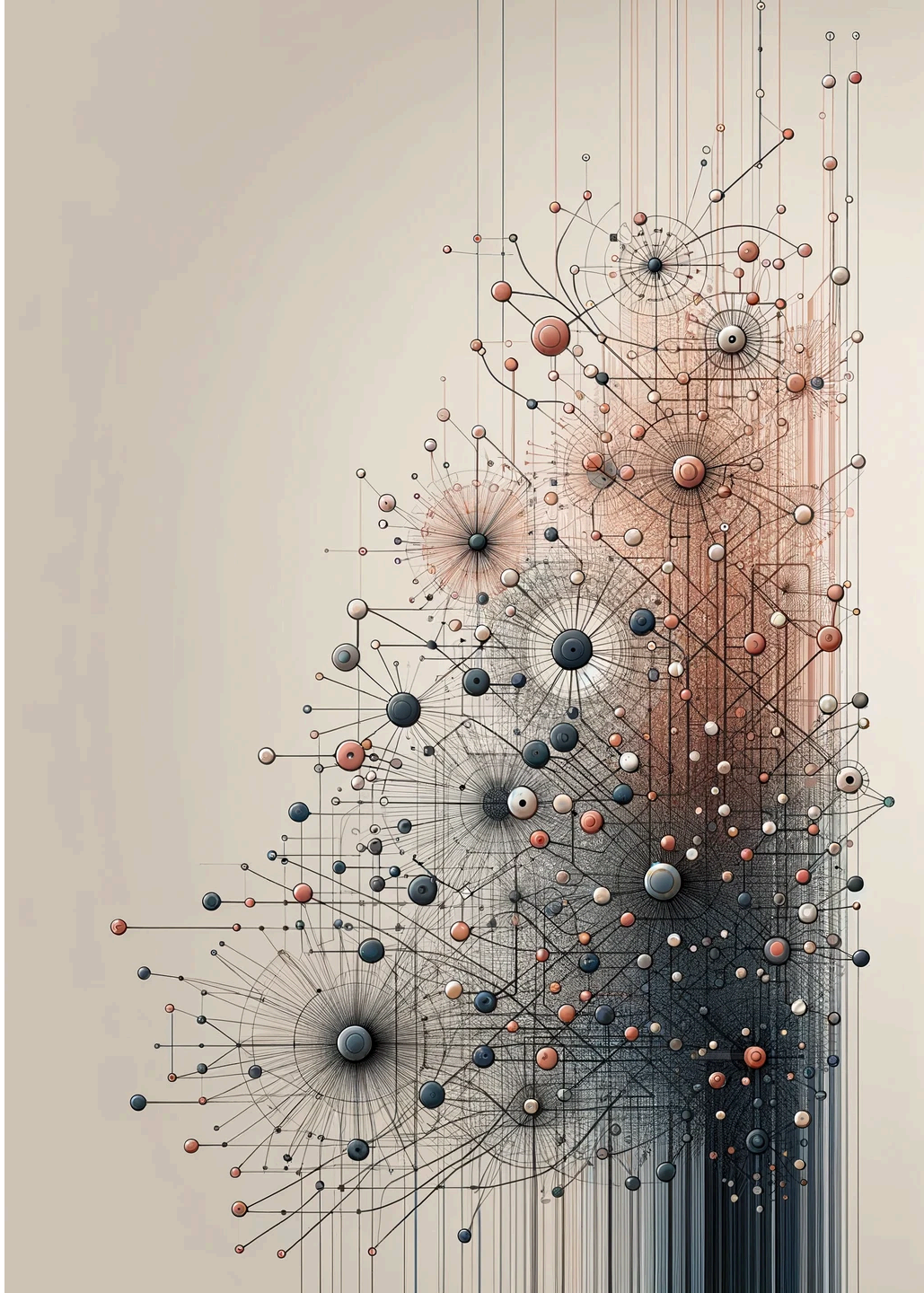
UNIVERSITÀ DI TORINO

Analisi e Visualizzazione delle Reti Complesse

NS16 - Games on Networks

Strategic Interaction and Equilibrium



Outline

- Why game theory belongs in network science
- Games, strategies, payoffs, and best responses
- Dominant strategies and the Prisoner's Dilemma
- Nash equilibrium and multiple equilibria
- Mixed strategies
- Pareto optimality and social optimality
- From games to traffic networks

From Networks to Strategic Systems

Structure tells us who can affect whom.

Game theory tells us how interdependent choices unfold on that structure.

From structure to behavior

So far, we mostly studied **network structure**:

- who is connected to whom
- which nodes are central
- how networks grow
- how communities organize mesoscopic structure

Now we ask a different question:

what happens when agents on or through a network make strategic decisions?

Why game theory in complex systems?

Complex systems are not only large collections of components.

They are systems where:

- local decisions interact
- each agent's outcome depends on other agents
- feedback can amplify small choices
- macro-level patterns emerge without central control

Game theory gives us a compact language for these mechanisms.

Why networks change strategic behavior

Networks shape strategic interaction because they define:

- **who interacts** with whom
- **which options** are available
- **where externalities** propagate
- **which equilibria** can be reached
- **how local incentives** aggregate into global outcomes

A game on a network is not just a game with many players.

The network is part of the mechanism.

Strategic Interaction

The first step is to describe decisions whose consequences depend on other decisions.

The complex-systems question

In this lecture, the central question is:

When many agents choose what is best for themselves, what happens to the system as a whole?

This question appears in:

- traffic congestion
- platform competition
- vaccination and herd immunity
- technology adoption
- climate agreements
- coordination and collective action

Independent choice versus strategic choice

An **independent decision** can be evaluated alone.

Example:

- choosing a book to read
- choosing a local file format
- choosing a personal study schedule

A **strategic decision** depends on what others do.

Example:

- choosing a road when others create congestion
- choosing a standard when compatibility matters
- choosing whether to cooperate when others may defect

What is a game?

A game is a simplified model of strategic interaction.

It specifies:

- the **players**
- the **strategies** available to each player
- the **payoff** each player receives for every combination of strategies

The key point is interdependence:

your payoff depends on your choice and on the choices of others.

Basic ingredients

1. **Players**

The decision makers.

2. **Strategies**

The possible actions for each player.

3. **Payoffs**

The numerical value of each outcome for each player.

These ingredients are often summarized in a **payoff matrix**.

What a payoff matrix represents

In a two-player game:

- rows are Player 1's strategies
- columns are Player 2's strategies
- each cell contains two payoffs
- the first payoff belongs to Player 1
- the second payoff belongs to Player 2

The matrix is a compact map from **joint behavior** to **individual outcomes**.

A first payoff matrix

Two classmates must choose where to meet, without communicating.

		Player 2	
		Library	**Cafe**
Player 1	**Library**	(3, 3)	(0, 0)
	Cafe	(0, 0)	(2, 2)

How to read one cell:

- if both choose **Library**, Player 1 gets 3 and Player 2 gets 3
- if they choose different places, both get 0

Bigger payoff means a more preferred outcome for that player.

Modeling assumptions

A payoff matrix is a model, not a full description of reality.

It usually assumes:

- players know the structure of the game
- players care about the payoffs shown in the matrix
- players try to choose strategies that improve their payoff
- choices are made simultaneously or without knowing the other's current move

These assumptions are simplifications, but useful ones.

What game theory is not saying

Game theory does not require that real people are perfect calculators.

It is a modeling language for asking:

- what incentives does the system create?
- what behavior is stable if agents respond to those incentives?
- where can local incentives conflict with collective welfare?

In complex systems, this is often enough to reveal the mechanism.

Running example: exam or presentation

Two students have two tasks due tomorrow:

- an **exam**
- a **presentation**

Each student must choose where to spend their effort.

They cannot communicate, and each student wants the best final score.

		Your Partner	
		Presentation	*Exam*
You	*Presentation*	$90, 90$	$86, 92$
	Exam	$92, 86$	$88, 88$

Reading the example

The example has two strategies:

- `E` : study for the exam
- `P` : prepare the presentation

The payoffs encode the final grade.

The important point is not the exact numbers.

The important point is that each student's best choice depends on what the other student does.

Best response

A **best response** is the best strategy against a fixed strategy of the other player.

If Player 2 chooses T, then strategy S is a best response for Player 1 if:

$$P_1(S, T) \geq P_1(S', T)$$

for every alternative strategy S'.

Best responses are the first step toward predicting behavior.

Strict best response

A **strict best response** is better than every alternative.

If Player 2 chooses T, then S is a strict best response for Player 1 if:

$$P_1(S, T) > P_1(S', T)$$

for every alternative strategy S'.

Strict best responses create stronger predictions because the player has no tie to resolve.

Dominant Strategies

Sometimes one action is best no matter what the other player does.

Dominant strategy

A strategy is **dominant** if it is a best response to every strategy of the other player.

A strategy is **strictly dominant** if it is a strict best response to every strategy of the other player.

High-level rule:

a dominant strategy is locally rational no matter what the other player does.

Exam-presentation: dominant strategy

In the exam-presentation example:

- each student has a strictly dominant strategy
- the dominant strategy is `E`
- the outcome is `(E,E)`

But `(P,P)` would give both students a better payoff.

This is our first warning:

individual rationality can produce a collectively worse outcome.

The Prisoner's Dilemma

Two suspects are interrogated separately.

Each can:

- confess
- not confess

The payoff is higher when the prison sentence is shorter.

		Suspect 2	
		NC	C
Suspect 1	NC	$-1, -1$	$-10, 0$
	C	$0, -10$	$-4, -4$

Prisoner's Dilemma: the mechanism

For each prisoner:

- if the other does not confess, confessing is better
- if the other confesses, confessing is still better

So confessing is strictly dominant for both.

The outcome is stable, but worse for both than mutual silence.

This is the canonical gap between **private incentives** and **collective welfare**.

Payoffs define the game

Small changes in payoffs can change the prediction.

If the reward structure changes, the dominant strategy may change.

This is why policies, norms, tolls, taxes, and incentives matter:

they change the game, not only the players.

		Your Partner	
		Presentation	*Exam*
You	*Presentation*	$98, 98$	$94, 96$
	Exam	$96, 94$	$92, 92$

One dominant player

Sometimes only one player has a dominant strategy.

Example:

- two firms choose between a low-priced product and an upscale product
- Firm 1 has a stronger brand
- direct competition favors Firm 1

		Firm 2	
		Low-Priced	*Upscale*
Firm 1	*Low-Priced*	.48, .12	.60, .40
	Upscale	.40, .60	.32, .08

Strategic reasoning with asymmetry

If Firm 1 has a strictly dominant strategy, Firm 2 can reason around it.

Firm 2 does not need a dominant strategy of its own.

It can ask:

if Firm 1 predictably chooses its dominant strategy, what is my best response?

Strategic reasoning often combines direct payoff comparison and prediction about others.

No dominant strategies

Many games have no dominant strategy.

In those cases, we cannot predict behavior by asking:

what should each player do regardless of the other?

Instead, we ask:

can we find strategies that are best responses to each other?

This leads to Nash equilibrium.

Example: three-client game

Two firms choose which client to approach.

The payoffs depend on:

- whether they approach the same client
- whether one firm can win a client alone
- whether a large client requires both firms

This example has no dominant strategy for either firm.

		Firm 2		
		A	B	C
Firm 1	A	$4,4$	$0,2$	$0,2$
	B	$0,0$	$1,1$	$0,2$
	C	$0,0$	$0,2$	$1,1$

Midpoint recap

We have three levels of reasoning:

- **best response**: best action against a fixed opponent choice
- **dominant strategy**: best action against every opponent choice
- **strategic prediction**: behavior must be mutually consistent

The next concept formalizes mutual consistency.

Equilibrium

When no one wants to move alone, a pattern of behavior can become stable.

Nash equilibrium: intuition

A **Nash equilibrium** is a stable pattern of strategies.

Stable means:

no player can improve their payoff by changing strategy alone.

This does not mean the outcome is morally good, socially optimal, or fair.

It means only that unilateral deviation is not attractive.

Nash equilibrium: definition

Suppose Player 1 chooses S and Player 2 chooses T.

The pair (S, T) is a Nash equilibrium if:

- S is a best response to T
- T is a best response to S

For many players, a Nash equilibrium is a list of strategies where every player is best-responding to all the others.

How to find Nash equilibria

For a two-player payoff matrix:

1. Fix each column and mark Player 1's best response.
2. Fix each row and mark Player 2's best response.
3. Look for cells where both players are best-responding.

Those cells are pure-strategy Nash equilibria.

This procedure is mechanical, but the interpretation is conceptual:

equilibrium means mutual best response.

Coordination games

Some games have multiple equilibria.

Example:

- two collaborators must choose software for a joint presentation
- either software works
- using the same software is much better than using different software

		Your Partner	
		PowerPoint	*Keynote*
You	*PowerPoint*	$1, 1$	$0, 0$
	Keynote	$0, 0$	$1, 1$

Multiple equilibria

Coordination games show that Nash equilibrium is not always a unique prediction.

Two outcomes may both be stable:

- `(PowerPoint, PowerPoint)`
- `(Keynote, Keynote)`

The game tells us which outcomes are self-reinforcing.

It may not tell us which one will be selected.

Focal points and conventions

Thomas Schelling emphasized **focal points**.

A focal point is a natural, salient equilibrium that players may converge on.

Examples:

- driving on the same side of the road
- using a shared language
- adopting a common technical standard
- meeting at a well-known landmark

Networked systems often rely on conventions to solve coordination problems.

Unbalanced coordination

Some coordination games have one equilibrium that is more attractive.

The payoff matrix may contain a clue about equilibrium selection.

But selection may also depend on history, expectations, institutions, and network position.

This is already a complex-systems idea:

path dependence can select among multiple stable states.

		Your Partner	
		PowerPoint	*Keynote*
You	*PowerPoint*	$1, 1$	$0, 0$
	Keynote	$0, 0$	$2, 2$

Network effects and coordination

Coordination games are especially important on networks.

Examples:

- choosing a messaging platform
- adopting a file format
- using a shared protocol
- coordinating behavior inside a community

The value of a strategy can increase when neighbors, partners, or peers choose the same strategy.

Hawk-Dove

Hawk-Dove is an anti-coordination game.

Two agents compete over a resource.

Each can behave:

- aggressively: `Hawk`
- passively: `Dove`

		Animal 2	
		D	H
Animal 1	D	$3, 3$	$1, 5$
	H	$5, 1$	$0, 0$

Hawk-Dove: equilibria

If one player is Hawk, the other prefers Dove.

If one player is Dove, the other prefers Hawk.

So the pure Nash equilibria are asymmetric:

- `(Hawk, Dove)`
- `(Dove, Hawk)`

The game is stable when roles differ.

Mixed Strategies

When no pure action is stable, strategic uncertainty can itself become stable.

Pure equilibria are not always enough

Some games have no pure-strategy Nash equilibrium.

In these games, every pure outcome gives at least one player a reason to switch.

To handle this, we enlarge the strategy space.

Players can randomize.

This leads to **mixed strategies**.

Matching Pennies

Two players choose heads or tails.

Player 1 wins if the choices differ.

Player 2 wins if the choices match.

There is no pure Nash equilibrium.

Every fixed choice can be exploited.

		Player 2	
		H	T
Player 1	H	$-1, +1$	$+1, -1$
	T	$+1, -1$	$-1, +1$

Mixed strategies

A mixed strategy assigns probabilities to pure strategies.

For Matching Pennies:

- Player 1 plays `H` with probability p
- Player 1 plays `T` with probability $1 - p$
- Player 2 plays `H` with probability q
- Player 2 plays `T` with probability $1 - q$

Pure strategies are special cases where the probability is 0 or 1.

Expected payoff

If Player 1 plays `H`, the expected payoff is:

$$(-1)q + (+1)(1-q) = 1 - 2q$$

If Player 1 plays `T`, the expected payoff is:

$$(+1)q + (-1)(1-q) = 2q - 1$$

Mixed strategies turn payoffs into expected values.

The indifference principle

In a mixed-strategy equilibrium, a player who randomizes must be indifferent between the pure strategies used with positive probability.

For Player 1 to mix between `H` and `T`:

$$1 - 2q = 2q - 1$$

Therefore:

$$q = \frac{1}{2}$$

Matching Pennies equilibrium

By symmetry, Player 2 mixes only if:

$$p = \frac{1}{2}$$

The mixed-strategy Nash equilibrium is:

$$p = q = \frac{1}{2}$$

The point is not that players literally flip coins.

The point is that predictable behavior would be exploitable.

Nash's existence result

John Nash proved that every finite game has at least one Nash equilibrium if mixed strategies are allowed.

This is a powerful result because:

- pure equilibria may not exist
- mixed equilibria guarantee a solution concept
- equilibrium becomes a general language for strategic systems

Existence does not imply uniqueness, efficiency, or easy computation.

Equilibrium and Welfare

The next question is not whether behavior is stable, but whether it is good for the system.

Pareto optimality

An outcome is **Pareto-optimal** if no other outcome can make someone better off without making someone else worse off.

Intuition:

there is no unexploited win-win improvement.

Important limitation:

Pareto optimality does not mean fairness.

An unequal outcome can be Pareto-optimal.

Pareto optimality in the running example

In the exam-presentation game:

- `(E,E)` is the dominant-strategy outcome
- `(E,E)` is not Pareto-optimal
- both students would be better off at `(P,P)`

So a stable strategic outcome can leave an obvious collective improvement unused.

This is one of the main reasons game theory matters for complex systems.

Externalities

An **externality** occurs when one agent's action affects another agent's payoff.

In networked systems, externalities are often transmitted through edges, routes, platforms, or shared resources.

They can be:

- positive, as in coordination and compatibility
- negative, as in congestion and overuse

Traffic is the next lecture's canonical negative externality.

Social optimality

An outcome is **socially optimal** if it maximizes total payoff:

$$\sum_i p_i$$

Social optimality asks:

which outcome is best for the system as a whole?

Nash equilibrium asks:

which outcome is stable under unilateral deviations?

These are different questions.

Comparing solution concepts

Concept	Question	Main limitation
Best response	What is optimal against this choice?	Local reasoning only
Dominant strategy	What is best regardless of others?	Often does not exist
Nash equilibrium	What is mutually stable?	May be inefficient
Pareto optimality	Are win-win improvements exhausted?	Says little about fairness
Social optimum	What maximizes total welfare?	May not be individually stable

Mechanism design intuition

If equilibrium is inefficient, one response is to change the game.

Examples:

- contracts
- norms
- taxes
- tolls
- subsidies
- platform rules
- regulation

The goal is to better align local incentives with system-level outcomes.

Route-choice teaser

Traffic is a perfect network game.

Each driver chooses a route.

The cost of a route depends on how many other drivers use the same roads.

Each driver can be individually rational.

The resulting traffic pattern can still be socially inefficient.

This is the bridge to `NS17`.

Final synthesis

Across the examples, the same pattern keeps returning:

System	Local rule	Collective issue
Prisoner's Dilemma	choose the dominant action	mutual loss
Coordination	match expectations	multiple equilibria
Hawk-Dove	avoid being exploited	asymmetric stability
Traffic	choose the fastest route	congestion externality

Main takeaway

Game theory gives us a language for complex systems with strategic agents.

The key lesson is:

stable local behavior is not the same as good global behavior.

In the next lecture, we will see this mechanism inside a networked infrastructure system:

traffic.

Reading material

Easley and Kleinberg, *Networks, Crowds, and Markets*

- Chapter 6: Games
- Sections 6.1-6.9

Recommended focus:

- best responses
- dominant strategies
- Nash equilibrium
- mixed strategies
- Pareto and social optimality

Q&A